

ShadowLab: A Windows-Focused Security Operations Platform

Ulfat Ibadov

github.com/ibadovulfat/shadowlab-detection-lab | shadowlab.about.surf

Overview

ShadowLab is a Python-based security operations and research platform built to study, simulate, and defend against offensive tradecraft on Windows systems. It runs on a FastAPI backend with a PySide6 desktop client and integrates host telemetry collection, process investigation, persistence discovery, threat intelligence enrichment, response orchestration, graph-based analysis, and enterprise case management into a single environment. It was built for detection engineers, threat hunters, and security researchers working on the problem of how attacks unfold and where detection pipelines fail against them.

Purpose and Research Focus

The platform addresses gaps that standard EDR tooling leaves open. One of them is evasion research: ShadowLab provides a structured environment to work through EDR bypasses, AMSI evasion, ETW manipulation, process injection, syscall proxying, unhooking, and parent process spoofing. Another is telemetry visibility: rather than collecting isolated events, the platform captures and correlates host signals across CPU, memory, threads, handles, files, and network connections so that operator behaviour shows up as a pattern rather than a series of disconnected data points. Detection itself is layered, combining external YARAify lookups, local YARA scanning across 1,163 enterprise rules, memory analysis, and a logistic regression model that produces fused behavioural verdicts in real time. Regression was the right choice for this context because forensic work requires verdicts that can be explained and defended, not just verdicts that are statistically accurate. The platform also handles enterprise investigation workflows through case management, role-based access control, task assignments, approval chains, and structured evidence handling.

Technical Stack

Backend	FastAPI, Uvicorn, SQLite/PostgreSQL, Python
Desktop Client	PySide6, pywin32, psutil, Watchdog
Detection Engine	YARA (1,163 rules), scikit-learn, Pandas, NumPy
Threat Intelligence	VirusTotal, MalwareBazaar, AbuseIPDB, YARAify
Visualization	Plotly, Matplotlib, Pyvis
Deployment	Docker, GitHub Actions CI/CD
Access Control	RBAC (viewer/analyst/admin), signed requests, policy enforcement

Open Research Questions

The platform works and has been tested against real adversary behaviour, but the questions it surfaces are not ones that more engineering can answer. What is the minimum behavioural signal that actually makes evasion detection reliable? If an adversary spends enough time studying how the model reaches its conclusions and starts working around it, is there a way to show the detection still holds, or does the confidence rest entirely on empirical testing? How do you specify a threat landscape formally enough to make guarantees about coverage rather than just observations about what has worked so far? Getting to answers requires formal threat modelling, adversarial machine learning theory, and verification methods that go beyond what independent research can provide, which is the core motivation for pursuing graduate study.

Future Direction

On the roadmap are self-tuning detection pipelines that adapt based on observed operator behaviour, container-native deployment for enterprise use, broader coverage across macOS and Linux, distributed tracing for platform observability, and tighter integration with OSSEC and WHIDS for active response. The longer-term goal is to move the behavioural scoring model from something that works empirically to something that can be formally verified, and to develop the platform into a research contribution to the literature on evasion-resistant detection.

Conclusion

ShadowLab was built on the premise that effective defence requires understanding offence at a technical level, not just in principle. By combining evasion research with detection engineering and enterprise response workflows, the platform enables iterative hardening of detection pipelines against the techniques that modern adversaries use. The work to date has produced a functioning system. The work ahead requires a formal theoretical foundation, which is what this research programme offers.